AegisOps

Evidence to action, with humans in control

- Manufacturing | Intelligent Operations & Industrial Efficiency

- For plant supervisors and reliability engineers: turn fragmented alarms and sensor readings into a traceable incident brief.

- Synthetic hackathon sandbox. No real equipment control or production savings claims.

The problem and the narrow wedge

One upstream fault should not become twenty tickets

- Fleet triage for plant equipment and its power chain, with chronological telemetry replacement.
- Group downstream supply symptoms without hiding independent bearing, pressure or battery risks.
- Product hypothesis: trained evidence, topology and a cited engineer handoff reduce gathering effort.
- Validate this hypothesis with an approved pilot; no invented customer or market statistics.

A complete, reviewable workflow

Monitor -> Diagnose -> Decide -> Act -> Verify

- Monitor: healthy-history robust-z + Isolation Forest, Markov event patterns and a live tool trace.
- Diagnose: topology correlation plus Vertex AI hypotheses and validated source passages.
- Decide/Act: reversibility policy, protected-asset gates, named sandbox approval or review-only handoff.
- Verify: two healthy future ticks within three; recurrence reopens. Uploads are never rewritten.
- 22-asset power demo: six synthetic scenarios, plus replaceable CSV/JSON manufacturing telemetry.

Google Cloud architecture

Real services, not decorative integrations

- Cloud Run: deployed application, scale-to-zero, one-instance demo limit.
- Gemini on Vertex AI: structured incident reasoning and work-order drafting.
- Cloud Storage: private SOP and synthetic scenario documents.
- BigQuery: partitioned audit table, 30-day retention, bounded history query.
- Dedicated runtime service account; no API key embedded in the cloud service.

Validation and honest limitations

Make failures visible; do not manufacture success

- Regression tests exercise approval, invalid output, bad citations, missing evidence, reopened cases, storage failure and rejected audit rows.

- Recorded live smoke test: 4/4 accepted Gemini diagnoses. This is not accuracy on a labeled production dataset.

- Quote checks establish textual provenance, not proof of every causal claim. Model confidence is self-reported.

- Assets, SOPs and post-action readings are synthetic. No production deployment or financial savings measured.

Impact, feasibility and the next step

Start with one approved workflow at one plant

- Pilot measures: manual-vs-assisted triage time, engineer acceptance, unsupported claims and appropriate escalation.

- Use approved site procedures, qualified industrial review, tenant isolation and authenticated access before any production pilot.

- Scale documents and audit across sites; add real monitoring ingestion only after validation.

- Future roadmap only: Pub/Sub, Dataflow and hosted predictive workloads.

- Submission: deployed demo + public source + this PDF + video strictly under three minutes.